GRA_Net

Deep Learning Model for Age and Gender Classification

Complete Student Project

Problem Statement

Age and gender estimation from unconstrained facial images is difficult because of pose, blur, illumination, expression and occlusion.

Goal: build a runnable deep-learning system inspired by the supplied GRA_Net paper.

Objectives

Implement gated residual attention.

Train age-category + age-unit models and a gender model.

Evaluate MAE and gender accuracy.

Deploy a Streamlit demo.

Paper Contributions

Classification + regression for age.

Separate gender classification.

Trainable gate added to Residual Attention Network.

Attention masks for feature selection.

Evaluation on five benchmark datasets.

Datasets

FG-NET: 1002 images, 82 subjects.

AFAD: 164,432 labeled photos in the paper.

Wikipedia Age: 62,328 images / 20,284 celebrities.

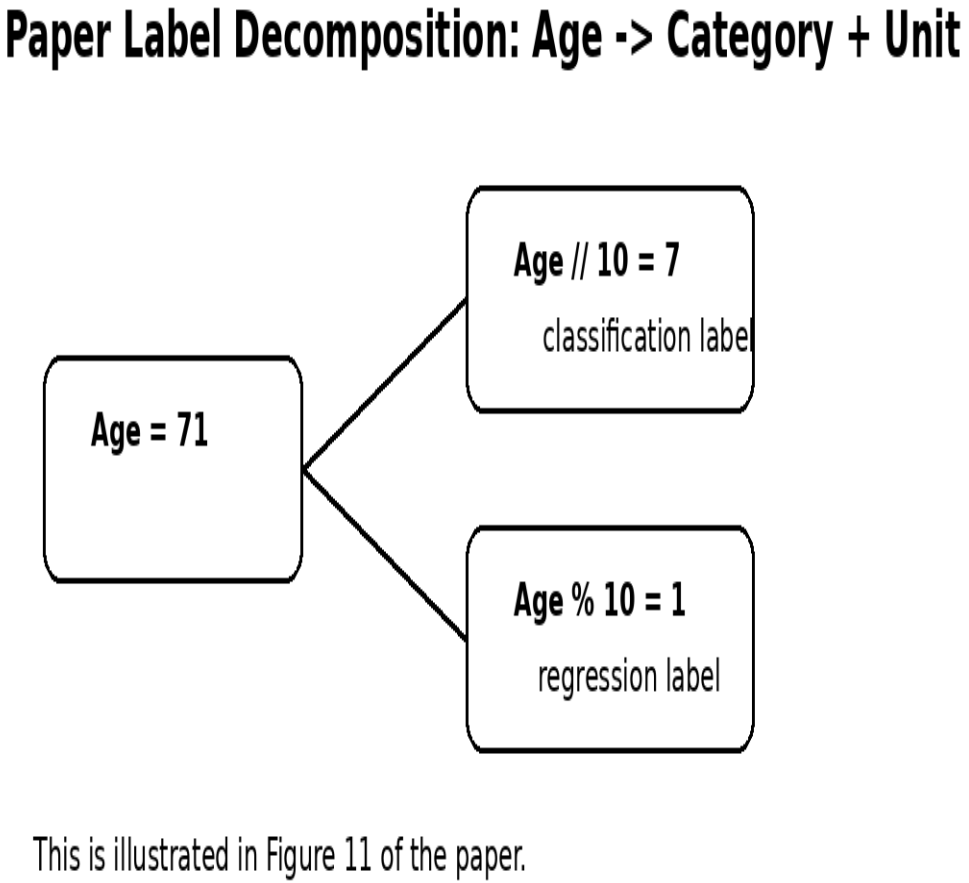
UTKFace: >20K images, age 0-116.

AdienceDB: Flickr-derived, manually labeled.

Age Label Decomposition

Example from Figure 11: age 71 -> age // 10 = 7 (classification) and age % 10 = 1 (regression).

Predicted age = predicted decade * 10 + rounded unit digit.



This is illustrated in Figure 11 of the paper.

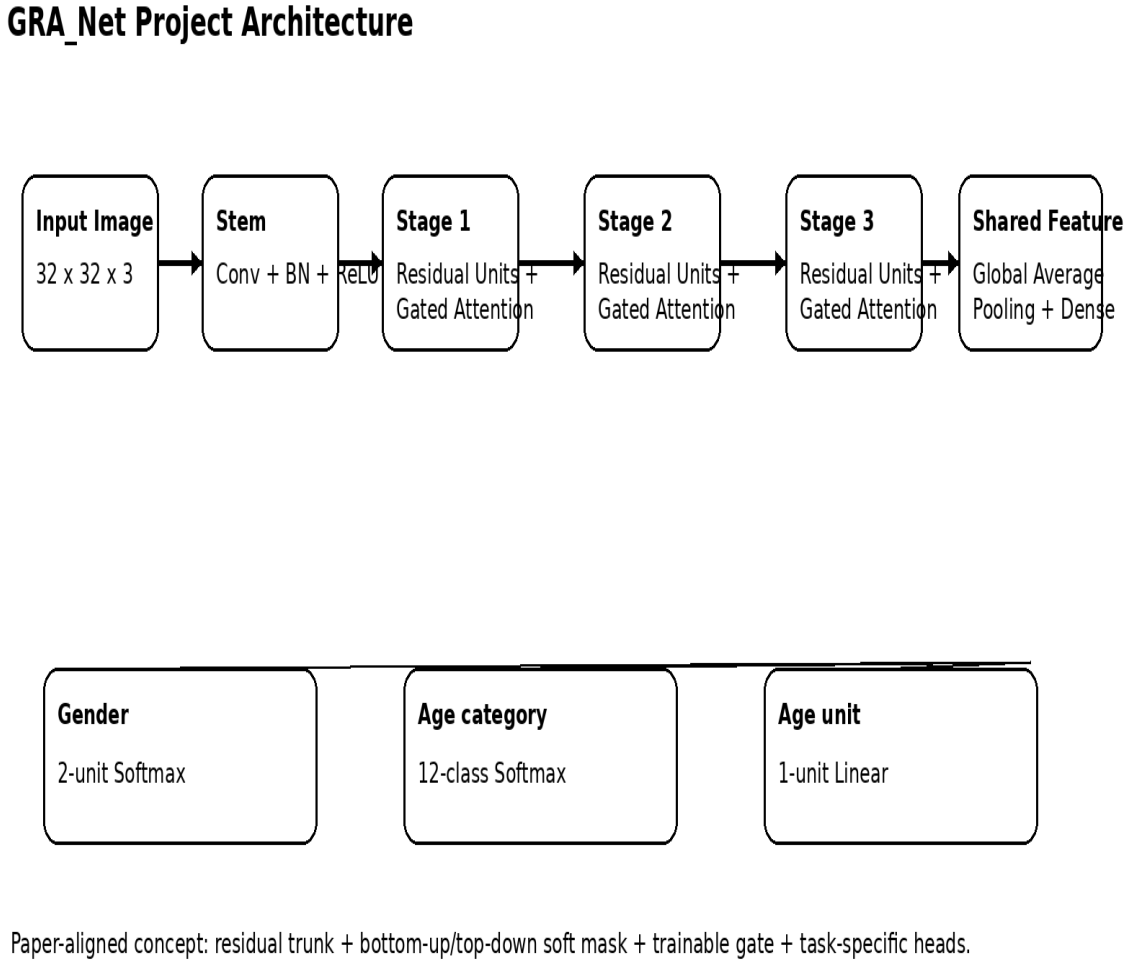
GRA_Net Architecture

Residual/ResNeXt-style trunk.

Bottom-up/top-down soft mask.

Trainable gated residual attention.

Three task-specific outputs in the student implementation.



Paper-aligned concept: residual trunk + bottom-up/top-down soft mask + trainable gate + task-specific heads.

Gated Residual Attention

Eq. (3): O = (gamma + (1-gamma)K) * F.

K is the soft mask; gamma is trainable.

The gate can preserve useful trunk features while allowing attention to modulate them.

Attention Modes

Mixed: sigmoid per channel and spatial position.

Channel: L2 normalization across channels.

Spatial: channel-wise standardization followed by sigmoid.

Training Settings

Input: 32x32 RGB.

Gender: 2-unit softmax + BCE.

Age category: categorical cross entropy.

Optimizer: Nadam, lr=0.001.

Age unit: single linear output; MSE is the project choice because the paper does not specify the regression loss in the experimental paragraph.

Evaluation

Primary paper metrics: age MAE and gender accuracy.

Project diagnostics: exact-age accuracy and within ±5 years.

Generate confusion matrix and age-error distribution.

Results - Fill After Training

Age MAE: [YOUR RESULT]

Gender accuracy: [YOUR RESULT]

Exact-age accuracy: [YOUR RESULT]

Within ±5 years: [YOUR RESULT]

Important: these must come from results/results.csv.

Paper Benchmark Reference

Reported by the supplied paper: FG-NET 3.23, AFAD 3.10, Wikipedia Age 5.45, UTKFace 1.07, AdienceDB 10.57 MAE.

Gender: 99.2% on UTKFace and 82% on AdienceDB.

These are literature reference values, not student results.

Error Analysis

Paper discusses blur, side pose, occlusion, glasses/hats, children and feature ambiguity.

Use your own test examples to explain where your trained model fails.

Deployment

Streamlit image upload.

Largest-face detection for the demo.

32x32 preprocessing.

Age + gender prediction and confidence display.

Project Structure

Core model: gra_net.py

Data: data_utils.py

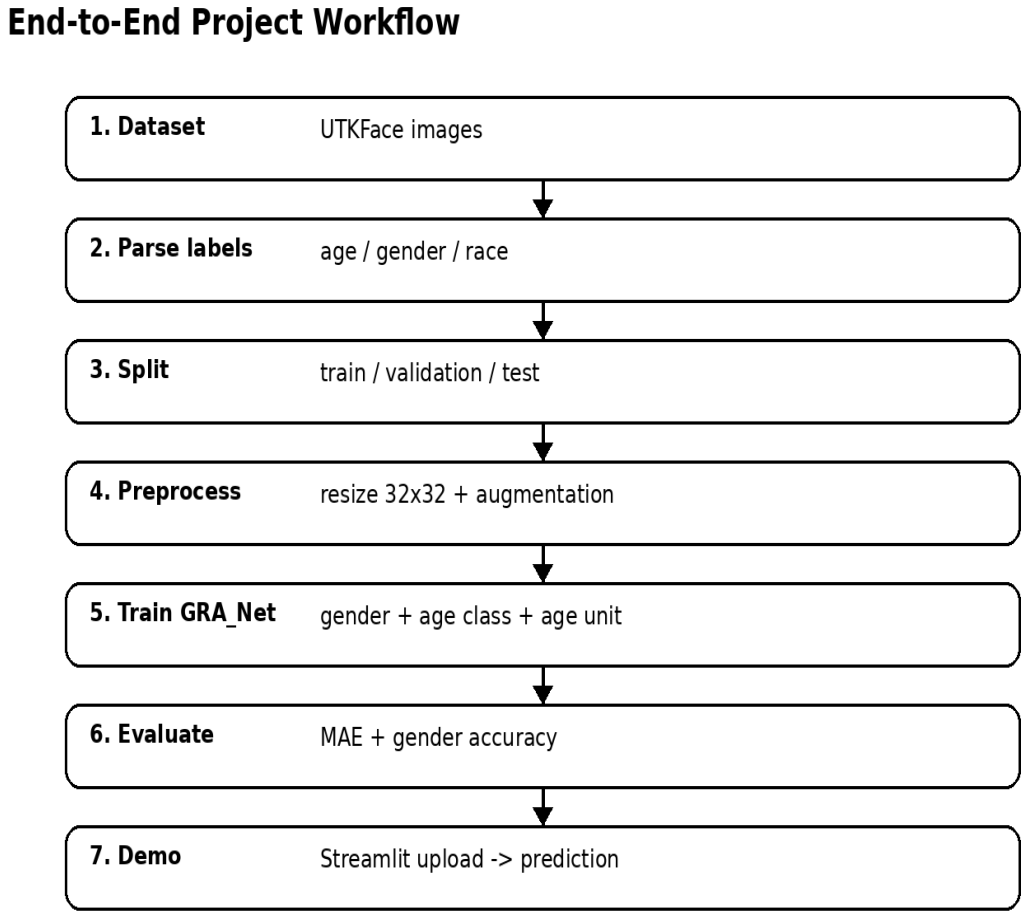
Training: train.py

Evaluation: evaluate.py

Inference: predict.py

Demo: app.py

Colab: notebooks/GRA_Net_Complete_Colab.ipynb



Limitations and Future Scope

Default UTKFace split is practical rather than subject-disjoint.

Compact stage configuration is a project choice, not a layer-by-layer paper reproduction.

Future: all five datasets, stronger occlusion/pose robustness, children, geographic robustness, improved losses.

Conclusion

A complete student pipeline was built from the supplied paper: preprocessing, GRA_Net, three task models, evaluation, inference and demo.

Always present your own experimental results separately from the paper benchmarks.